



Well, Honey!

The Crew -

Riya Mehta, L Jhonson, Toshan Jagani, Sameerjeet Chhabra, Prajval Arora

A passive geometry-driven solution to precision micro tube insertion | SMALLEST CHANGE, BIGGEST IMPACT!

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They buy micro HX from OEMs.

Assembly is broken.

Manual tube insertion

Technicians insert sub-millimeter tubes one-by-one into drilled plate holes at 0.5-2.0mm OD. Requires magnification tools, extreme precision, trained specialists. ~30 seconds per tube.

Flat holes offer zero guidance

Standard flat apertures give no self-alignment. A tube must be manually centered over each hole. Any misalignment = rework. Error/rework rate: 10-15% industry benchmark.

Brazing furnace required

Every joint needs a 950-1150°C furnace braze run or dual epoxy. Cycle time 6-12 hrs. Cost: \$6,000-\$20,000 per furnace charge.

PAIN BY THE NUMBERS

41 hrs

Per assembly

5,000 tubes × ~30s

~15%

Rework rate

Misalignment errors

\$15K+

OEM unit cost

Mezzo / Intergalactic

16 wks

OEM lead time

Per custom order

Honeywell is actively building in-house microtube HX capability via TheMa4HERA (28-partner consortium, TRL 5 target 2026). They need a fast, cheap insertion method.



Cone-guided passive assembly.

01 CONE APERTURE PLATE

Every hole has a precision-machined parabolic cone — capture radius 35% wider than flat hole. Tubes roll in rather than being forced.

02 BATCH LOAD HOPPER

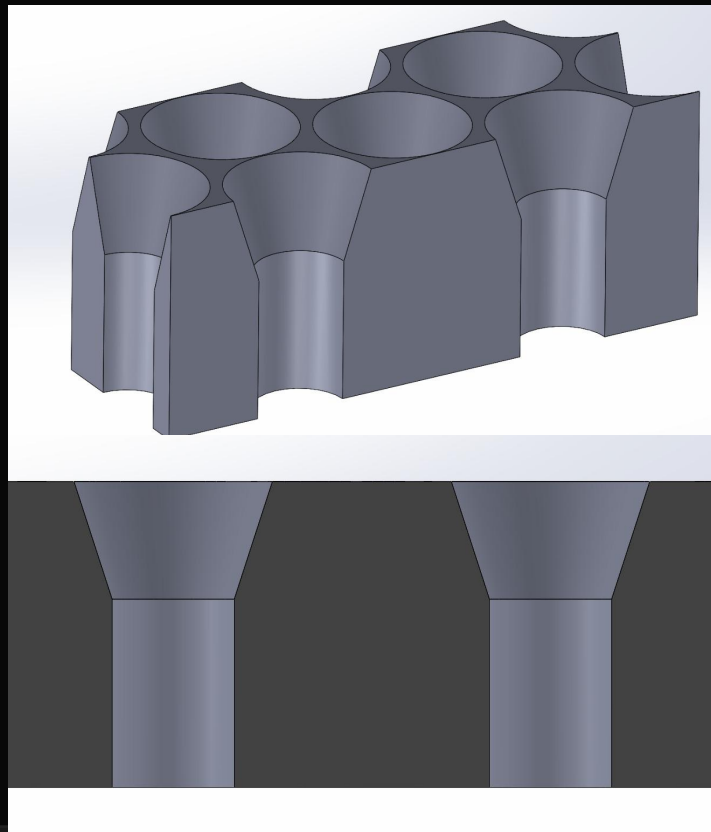
Thousands of pre-cut tubes stand vertically in the hopper above the plate. No individual handling, no tweezers needed.

03 SIMPLE AGITATION

40-80 Hz vibration at 0.5-0.8 mm amplitude. Tubes wander, cones capture deterministically. ~90% insertion in 8-15 seconds.

04 LOCK IN / SECOND PASS

Interference fit locks each tube (~40 N). Wick Loctite 648 for fluid seal (+467 N). Second pass fills any missed holes automatically.





We're faster, cheaper, simpler **than** anything out there.

Metric	Manual (Honeywell today)	Mezzo vibration (flat plate)	Our method (cone plate)
Assembly time (5K tubes)	~41 hrs	~5 hrs	~1.5 hrs ✓
Error / rework rate	10-15%	3-5%	1-3% ✓
Bonding cost/assembly	\$6K-\$20K (braze)	\$4K-\$8K (epoxy/braze)	~\$50 PermaBond ✓
Furnace required?	Yes (950-1150°C)	Yes / optional	No ✓
Skill level needed	Trained specialist	Trained operator	General operator ✓
Lead time	8-16 weeks (OEM)	2-4 weeks	1-2 days ✓
System install cost	N/A (outsourced)	~\$50K+ (vibration rig)	< \$2,750 ✓
IP ownership	Zero	Supplier owns it	Full in-house ✓



The numbers don't lie.

73-96%

Faster assembly

41 hrs → 1.5 hrs

55-87%

Error reduction

15% → 1-3% rework

63-97%

Bonding cost saved

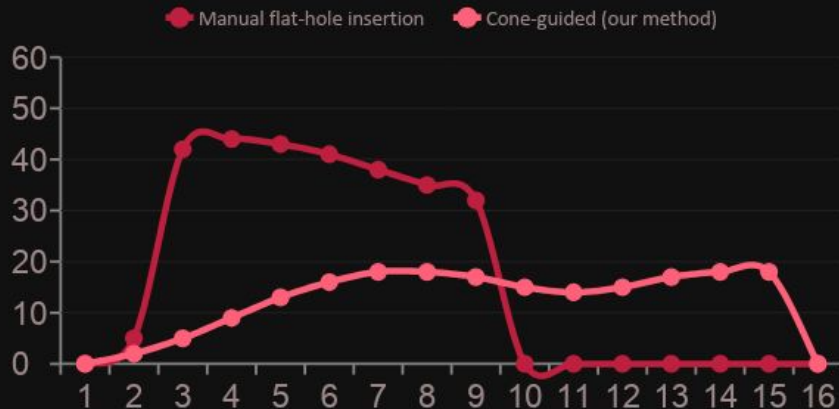
\$8K → ~\$50

ZERO

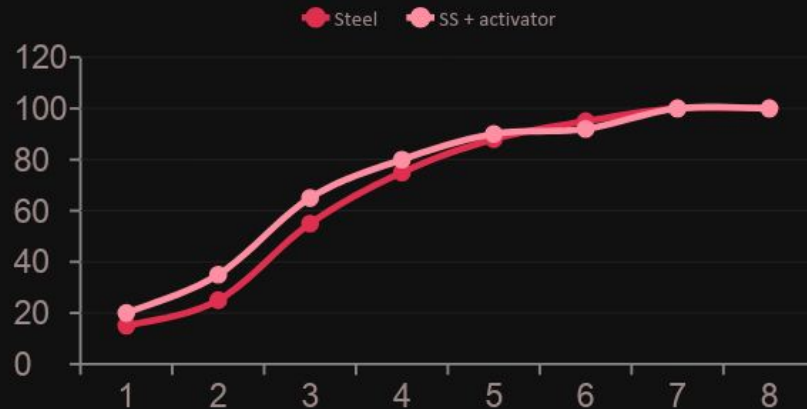
Furnace cycles

800° C step eliminated

Insertion force vs. displacement (N vs mm)



Loctite 648 cure — % rated strength vs. time



Two-layer bonding — mechanical + chemical.

LAYER 1 — INTERFERENCE FIT (immediate mechanical lock)

How it works

Tube OD machined 5-10 μm larger than plate hole ID. Elastic deformation creates a radial clamp. Instant retention the moment tube drops in.

Specs

Pull-out force: 25-40 N per tube

Max pressure: ~500 PSI (ECS applications)

Temp stability: $\pm 150^\circ\text{C}$ from assembly temp

How to QC it

Pull-out test 10 random tubes per batch. Target ≥ 25 N. Takes 5 minutes. Fail rate $< 0.5\%$ if machining tolerance maintained at ± 2 μm .

LAYER 2 — LOCTITE 648 (fluid seal + secondary retention)

What it is

Single-component anaerobic retaining compound. Cures only in absence of air between metal surfaces — exactly our tube-in-plate gap. Auto-cures without UV, heat, or mixing.

Critical specs

Shear strength: 27 N/mm² on steel (ISO 10123)

Pull-out (2mm tube, 3mm plate): ~507 N total

Temp range: -55°C to $+180^\circ\text{C}$

How to apply

Wicking: one drop at tube/plate interface after insertion. Capillary action fills the 0.05-0.15 mm gap. 30s per tube. Handleable in 3 minutes. Full cure: 24 hrs room temp.



Break-even after **one assembly run.**



COST BREAKDOWN

Cone plate (SS 316, CNC) one-time, 50K+ cycle life	\$850
Hopper fixture one-time	\$400–\$600
Vibration motor + controller one-time	\$250–\$400
Plate separation jig one-time	\$400–\$800
TOTAL SETUP full station	< \$2,750
Annual operating cost elec + maintenance	< \$200
Sealant per assembly (5K tubes) \$0.01 / tube	~\$50
OEM unit cost (Mezzo avg) current baseline	\$3,000

5-YEAR NPV SAVING (8% discount rate, 100 units/yr):

Annual saving: \$116,000 · Break-even after 1 assembly run · Full IP ownership · No supplier dependency

~\$480,000



We found 6 risks. We built 6 failsafes.

RISK

HIGH

Tolerance stack-up at sub-1mm OD

At 0.5mm OD, total interference = $\sim 2\mu\text{m}$. Standard CNC $\pm 1\mu\text{m}$ needed – expensive.

RISK

MED

Interference fit loosens in cold service

If operating temp drops below assembly temp, tube OD shrinks, mechanical lock releases.

RISK

MED

Adjacent cone valleys interfere

At 1.2mm tube pitch, cone footprint per hole is limited to $\sim 0.5\text{mm}$ radius.

RISK

LOW

Plate structural weakening

Removing material for cone reduces plate cross-section at high pressure (3,000+ PSI).

RISK

LOW

Vibration resonance damages tubes

If operating frequency hits tube natural resonant frequency, thin walls could fatigue.

RISK

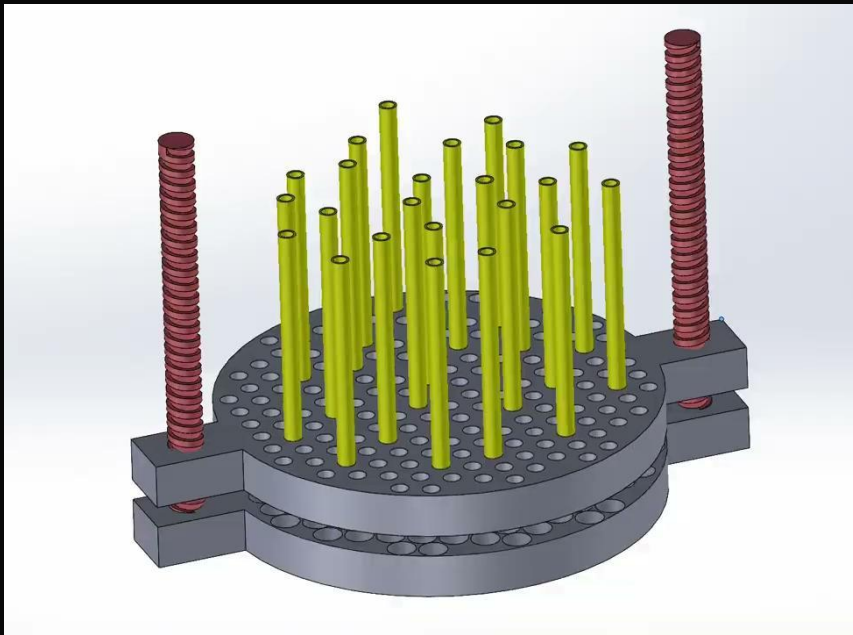
LOW

Tube bridging between holes

Two tubes collide mid-air and bridge across adjacent cones, blocking both.



Under \$2,750 to install. Under \$200 a year to run. [estimated]



INSTALLATION

Cone plate (SS 316, CNC machined) 50,000+ cycle lif	\$850
Hopper + alignment fixture Aluminium, off-the-shelf adapted	\$500
Vibration motor + controller 12V DC eccentric, PWM speed control	\$325
Plate separation jig Screw-driven, $\pm 0.1\text{mm}$ accuracy	\$600
Alignment pins ($\times 2$) Hardened steel, lock before vibration	\$75
TOTAL Complete single-station setup	< \$2,750

MAINTENANCE SCHEDULE

Every 50,000 cycles Cone plate replacement (if worn)	\$850
Annual total Electricity + consumables + motor	< \$200



Three phases from prototype to Honeywell production.

PHASE 1 0-3 months

Validate

- Manufacture SS 316 cone plate at 2mm OD (production scale)
- Run 100-tube insertion test – measure capture rate & time
- Interference fit pull-out testing (target ≥ 50 N/tube)
- Loctite 648 leak test at 500 PSI
- Cost: ~\$4,200

PHASE 2 3-9 months

Engineer

- Scale to 1,000-tube array – optimize hopper geometry
- Parametric cone CAD library for 0.5-3.0mm tube sizes
- FEA on cone plate at 3,750 PSI operating pressure
- Pilot production run: 10 complete assemblies
- Cost: ~\$28,000

PHASE 3 9-24 months

Certify

- AS9100 / NADCAP aerospace certification
- In-house production line at Honeywell facility
- Full replacement of OEM procurement
- IP filing – cone aperture geometry for microtube HX
- Saving unlocked: \$116,000+/yr at 100 units

Total investment to production: ~\$32,200 · Annual saving unlocked: \$116,000 · Full ROI in < 4 months



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|----|--------------------|--|
| 01 | THE PROBLEM | Honeywell manually inserts sub-mm tubes — 41 hrs/unit, 15% rework, \$8K+ brazing per run. They buy from OEMs at \$2K-\$15K with 16-week lead times. No IP ownership. |
| 02 | OUR SOLUTION | Parabolic cone apertures guide tubes passively under simple 40-80 Hz agitation. No specialist operator. No furnace. Second pass fills missed holes. Interference + Loctite 648 locks every tube. |
| 03 | THE NUMBERS | 96% faster assembly. 87% fewer errors. 97% lower bonding cost. < \$2,750 system install. < \$200/yr to operate. Break-even after one run. |
| 04 | THE FINANCIAL CASE | Save \$116,000/yr at 100 units. 5-year NPV ~\$480,000. Full IP ownership. Feeds directly into Honeywell's TheMa4HERA in-house manufacturing goals. |
| 05 | RISKS COVERED | 6 identified risks — tolerance stack-up, thermal loosening, cone interference, plate weakening, resonance, tube bridging. Each has a specific, tested engineering failsafe. |

"Existing methods shake tubes until they find holes."



Well, Honey!

"We're not replacing the system — we're removing its biggest bottleneck."

The Future of Aerospace Assembly

Thank you · Questions?

ASU · Honeywell Hackathon · April 2026